

CT2: CSE 201 - Object-Oriented Programming I: Java (Sec C)

Time: 30 min

Mark: 20

Name:

ID:

- Q1. Create a class named “**Contestant**”. Add the following to the class. [9]
- a. Attributes: **teamName**, **noOfProblems**, **noOfSolvedProblems**. Make the **noOfProblems** variable **static** and **int** type, also assign a value 10.
 - b. Constructor: A parameterized constructor and pass parameter for **teamName** and **noOfSolvedProblems**. Inside the constructor initialize the respective attributes using the “**this**” keyword.
 - c. Methods:
 - i. **public void solveAProblem()** – Increase the **noOfSolvedProblems** by 1.
 - ii. **public int getScore()** – return the **noOfSolvedProblems** attribute.
 - iii. **public void display()** - print the value of all attributes.
- Q2. Create a class “**ProgrammingContest**” and add main method. Inside the main method, add the following. [9]
- a. Create an object of the above class “**Contestant**” and pass your name as the teamname and (last digit of your id%3 + 1) as no of problems solved. Assign the reference of the object to a variable name “**team1**”.
 - b. Call method **solveAProblem()** method twice using the **team1** variable.
 - c. Call the **getScore()** method using the **team1** variable and print the value return by this method.
 - d. Call the display method.
 - What is the output of this call?
- Q3. a. Declare and create an array of size 10 and of **Contestant** type. [2]

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- Q1. Create a class named **"Player"**. Add the following to the class. [9]
- a. Attributes: **name, sport, noOfMatchesPalyed, and noOfMatchesWon**. Make the **noOfMatchesPalyed** variable **private** and of **int** type.
 - b. Constructor: A parameterized constructor with parameters for all attributes and inside the constructor initialize the respective attributes using the **"this"** keyword.
 - c. Methods:
 - iv. **public void play(boolean hasWin)** – Increase the **noOfMatchesPalyed** by 1. If the value of **hasWin** is true, increase the **noOfMatchesWon** by 1.
 - v. **public int getRank()** – return the **noOfMatchesWon** attribute.
 - vi. **public void display()** - print the value of all attributes.
- Q2. Create a class **"Tournament"** and add main method. Inside the main method, add the following. [9]
- a. Create an object of the above class **"Player"** and pass your name, **"Basket Ball"**, last 2 digit of your registration id and first digit of your registration number to the constructor as name, sport, noOfMatchesPlayed and noOfMatchesWon. Assign the reference of the object to a variable name **"player1"**.
 - b. Call method **play(...)** method using **player1** variable and pass true if your id is even and false for odd.
 - c. Call the **getRank()** method using the **player1** variable and print the value return by this method.
 - d. Call the display method.
 - What is the output of this call?
- Q3. a. Write an import statement to import a class name **Quiz** which is under **cse203.secD** package. [2]

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- Q1. Create a class named **"Publication"**. Add the following to the class. [8]
- Attributes: **title, author, citation** (should be int type) and make the **citation** variable **private**.
 - Constructor: A parameterized constructor with parameters for all attributes and inside the constructor initialize the respective attributes using the **"this"** keyword.
 - Methods:
 - public void citePublication()** – inside the method increase the **citation** by 1.
 - public int getCitation()** – return the **citation** attribute.
 - public void display()** - print the value of all attributes.
- Q2. Create a class name **Journal** and add main method. Inside the main method, add the following. [8]
- Create an object of the above class **"Publication"** and pass your **"OOP CT2 modeling"**, your name, and last 2 digits of your id to the constructor as the title, author and citation. Assign the reference of the object to a variable name **"oop"**.
 - Call method **citePublication()** twice using the **oop** variable.
 - Call the **getCitation()** method using the **oop** variable and print the value return by this method.
 - Call the **display()** method.
 - What is the output of this call?
- Q3. a. Declare a **package x.y** where **x**=your home town and **y**=your first name. 2+2
- b. Assume there is a class **"Person"** under this package x.y (of previous question) and class **"Test"** under package x where x =your home town. Write one statement that will allow you to access **Person** class from **Test A**.

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- Q1. Create a class named **"CR_Candidate"**. Add the following to the class. [8]
- Attributes: **name, age, noOfVotes** (should be int type) and make the **noOfVotes** variable **private**.
 - Constructor: A parameterized constructor with 2 parameters for name and age and inside the constructor initialize the respective attributes using the **"this"** keyword.
 - Methods:
 - public void votedFor(int noOfVoter)** – inside the method increase the **noOfVotes** by **noOfVoter** amount.
 - public int getNoOfVotes()** – return the **noOfVotes** attribute.
 - public void display()** - print the value of all attributes.
- Q2. Create a class **"VotingCentre"** and add main method. Inside the main method, add the following. [8]
- Create an object of the above class **"CR_Candidate"** and pass your name, age as the parameter of the constructor. Assign the reference of the object to a variable name **"candidate1"**.
 - Call method **votedFor(...)** method using **candidate1** variable and pass last digit of your registration number as the parameter
 - Call method **votedFor(...)** method using **candidate1** variable and pass first digit of your registration number as the parameter
 - Call the **getNoOfVotes()** method using the **candidate1** variable and print the value return by this method.
 - Call the display method.
 - What is the output of this call?
- Q3. a. Declare a **String** type instance variable named **"password"** such that no one can access this variable outside the class. 2+2
- b. Declare a **double** type instance variable named **"salary"** such that no one from a different package can get access to this variable.

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- Q1. Create a class named **"Batsman"**. Add the following to the class. [8]
- d. Attributes: ***name, rank, noOfCentury*** (should be int type) and make the ***rank*** variable **private**.
 - e. Constructor: A parameterized constructor with parameters for all attributes and inside the constructor initialize the respective attributes using the **"this"** keyword.
 - f. Methods:
 - vii. ***public void play(int score)*** – if the score is greater than 100, for every 100 score increase the ***noOfCentury*** by 1 and rank by 0.5.
 - viii. ***public int getRank()*** – return the ***rank*** attribute.
 - ix. ***public void display()*** - print the value of all attributes.
- Q2. Create a class **"Tournament"** and add main method. Inside the main method, add the following. [8]
- e. Create an object of the above class **"Batsman"** and pass your name, last 2 digit of your registration id and first digit of your registration number to the constructor as name, noOfCentury and rank. Assign the reference of the object to a variable name **"batsman1"**.
 - f. Call method ***play(...)*** method using ***batsman1*** variable and pass last 3 digits of your registration number.
 - g. Call the ***getRank()*** method using the ***batsman1*** variable and print the value return by this method.
 - h. Call the display method.
 - What is the output of this call?
- Q3. b. Declare and create an array of size 10 and of **Quiz** type. 2+2
- c. Write an import statement to import a class name **Quiz** which is under **cse203.secD** package.

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- Q1. Create a class named **"Library"**. Add the following to the class. [8]
- d. Attributes: ***name, address, totalBooks***. Make the ***totalBooks*** variable an int type and **private**.
 - e. Constructor: A parameterized constructor with parameters for all attributes and inside the constructor initialize the respective attributes using the **"this"** keyword.
 - f. Methods:
 - iv. ***public void addBooks(int quantity)*** – inside the method increase the ***totalBooks*** by ***quantity*** amount where ***quantity*** is the parameter of the method.
 - v. ***public int getTotalBookCount()*** – return the ***totalBooks*** attribute.
 - vi. ***public void display()*** - print the value of all attributes.
- Q2. Create a class name **University** and add main method. Inside the main method, add the following. [10]
- e. Create an object of the above class **"Library"** and pass "JRC memorial", your address, and last 4 digits of your id to the constructor as the name, address and totalBooks. Assign the reference of the object to a variable name **"uapLib"**.
 - f. Call method ***addBooks()*** using the ***uapLib*** variable and pass last 2 digits of you id as the quantity.
 - g. Call the ***getTotalBookCount()*** method using the ***uapLib*** variable and print the value return by this method.
 - h. Call the ***display()*** method.
 - What is the output of this call?
- Q3. c. Declare a package named **x.y** where **x="uap"** and **y=your first name**. [2]

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- Q1. Create a class named **"Appliance"**. Add the following to the class. [9]
- d. Attributes: **name, model, isOn, power** (power is in KWH unit). Make the **isOn** variable **private** and **boolean** type.
 - e. Constructor: A parameterized constructor with 3 parameters for name, model and power. Inside the constructor initialize the respective attributes using the **"this"** keyword.
 - f. Methods:
 - iv. **public void turnOn()** – inside the method set the **isOn** to **true**.
 - v. **public double run(double hour)** – return the power consumption which **power*hour**.
 - vi. **public void turnOff()** - inside the method set the **isOn** to **false**.
 - vii. **public void display()** - print the value of all attributes.
- Q2. Create a class **"Home"** and add main method. Inside the main method, add the following. [9]
- f. Create an object of the above class **"Appliance"** and pass **"AirCooler"**, **"Walton"**, and **0.2** as the parameter of the constructor. Assign the reference of the object to a variable name **"refg"**.
 - g. Call method **turnOn()** method using **refg** variable.
 - h. Call method **run(...)** method using **refg** variable and pass the (last digit of your registration + 2) as the parameter
 - i. Call the **display** method.
 - What is the output of this call?
- Q3. a. Declare a **double** type instance variable named **"salary"** such that no one from a [2]
different package can get access to this variable.